CSC15S Assignment 7 – Security Design Principles and Linux Process Management

Total: 100pts (due date in Blackboard)

What to turn in: upload the following to **Assignment 7** in Blackboard. (1) CSC15Slab7.docx (with all questions completed) (2) log_Mission5.txt

Part 1. Security Design Principles.

[20 pts. Estimated time to complete: 15 minutes.]

Read about all the secure design principles and list ONE and **ONLY ONE** design principle that fits the **BEST** for each of the following scenarios.

Case Study	Applicable Security Principle
Case 1: Docker Container Similar to virtual machines, many cloud platforms use a new technique called "docker container". Each individual application service (e.g., Apache and MySQL) is contained in one docker container and each enjoys a "logically" separate operating system. So after compromising one service (e.g., Apache), a hacker will still have difficulty at hacking another service (e.g., MySQL) because they are "logically" separated.	Isolation
Case 2: Linux Kernel Implementation If you compare the source code of Linux kernel (e.g., 2.x vs 4.x), the internal implementation has changed a lot (many due to the need to fix vulnerabilities)! But if you are an application programmer, you do not feel much the change – the API interface is pretty much the same. If you are using `write(fid, buf, len)` to write to files 40 years ago, you are still doing the same thing in your C program.	Abstraction
Case 3: Linux Kernel Modules As computer hardware evolves, many drivers may emerge after an operating system kernel has been stable (i.e., security vulnerability and bugs fixed). Linux allows new device drivers be dynamically loaded as a module so that the core kernel does not have to be changed.	Modularity
Case 4: Docker vs Full Fledged Virtualized OS Although virtualization has been the main drive of hosting services on cloud platforms, docker based technology has been the recent trend. Consider hosting an Apache web server, using virtualized OS you may have to install a full-fledged Cent-OS which has many unnecessary drivers and services installed; but the docker container will strip away unneeded kernel modules and generate a minimal Linux kernel for your application service. Which one is more secure? It's not hard to tell.	Minimilization
Case 5: Intercontinental Ballistic Missile (ICBM) Control To launch ICBM for planting nuclear mushrooms requires 8 steps in United States, involving many personnel. For example, the president issues orders via multiple layers of control. At the ICBM site, at least two officers (geographically located at least miles away from each other), independently receive command codes. Only when both officers verified the command codes, and both turned on the launch key, can the ICBM be fired.	Separation of Duties

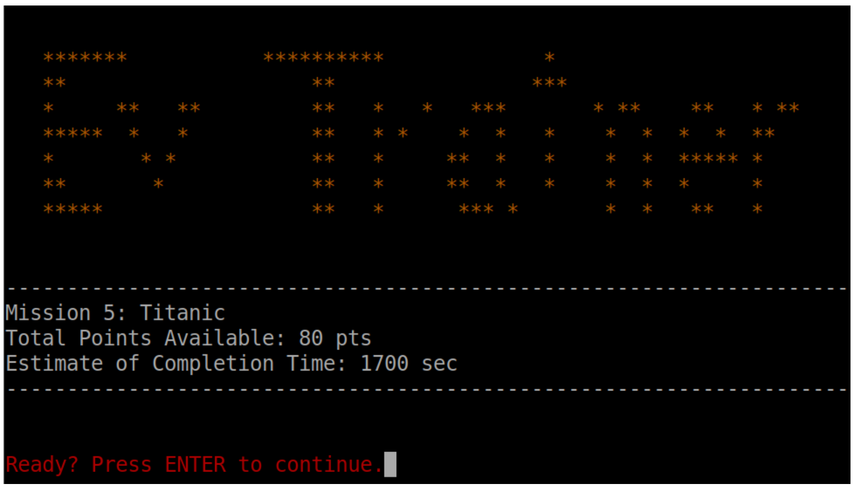
Part 2. Dr. Evil's Training Room. Mission 5 – Titanic

[80 pts. Estimated time to complete: 1.5 hrs]

This assignment is created in memory of the 1997 movie "Titanic". Dr. Evil hates tragedies. You are sent by Dr. Evil as a time traveler, carrying a laser gun. Your job is to kill evil icebergs (processes in your system) with the laser gun. Save Jack and Roses's love! Follow the instructions below to extract the related files:

```
tar -xvf mission_titantic.tar
cd mission_titanic
chmod 755 *
sudo ./mission_titanic
```

As usual, you need to start **another terminal** to complete the training. Once you feel ready to submit, please attach **log_Mission5.txt** to Blackboard. The following are some additional hints for each of the challenges.



Challenge	Hint
1. Sailing Log	Q1: Pay attention that the file ids for stdout and stderr are 1 and 2, respectively. Q2: Grep on alert.txt. The answer should be some number greater than 900.
2. Heart of Ocean	Q1: Either pipe or file redirection will work. The response is a song that everybody sings when (s)he is a toddler.
3. Jack's Innocence	Q1. Jack opened one and only one object. Find it out using strace.
4. Watchstander	Q1: Use the '*' well for crontab. Run it as root. Make sure that you are using **absolute path** for the command. If there is anything wrong, check /var/log of crontab.
5. Save Titanic	Q1: Use either "-eo" or "--sort" of the ps command to show VSZ and RSS. Q2: You got to act quick. Use ps sort to find out the PID of the bad process. Note: to kill a root process, you need `sudo kill -9 pid`. Q3: I find top quite handy in determining CPU intensive processes. Q4. The key is to use '*' wildcard match in your regular expression.